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Interfaces with Other Disciplines

Entropy in good manufacturing system: Tool for quality assurance

Pradeep K. Jha [a,b,c,*], Rakhi Jha [a], Rajul Datt [d], Sujoy K. Guha [a,*]

[a] School of Medical Science and Technology, RISUG[®] Pilot Plant, Indian Institute of Technology, Kharagpur 721302, India
[b] Department of Management Science, Uttar Pradesh Technical University, Lucknow 226021, India
[c] RISUG R&D Laboratory, Science and Technology Entrepreneurship Park, Indian Institute of Technology Kharagpur 721302, India
[d] J.P. School of Business, Meerut Cantt 200001, India

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ABSTRACT

It has been customary to implement Good Manufacturing Practices (GMP) in pharmaceutical organizations as a systematic and comprehensive quality approach and sometimes by regulatory enforcement. In this scenario, determination of the obvious entropy/disorder arising during the implementation has not been taken care of yet. Therefore, this paper gives the basis for applying query and visual perception of GMP system driven visualization approach, particularly the Laplace equation, to the determination of disorder and deviation pattern inside the GMP system applied in the organization. In this study, a three-dimensional mesh approached with raw and intermediate input handled under GMP parameter is considered to produce high quality products with minimum entropy (variation distribution) by adding the analogy wise different GMP parameters and process variables with Gauss Seidel iteration and thus producing visual picture of the entire system. The approximation involved in applying the equations to the GMP compliant aseptic region was analyzed. Using numerical technique and computer program, the Gauss Seidel iteration equations have been solved with appropriate GMP parameter and process variable. The result indicates that deviations vary over the GMP compliant system and that the process entropy affects the totality of disorderness. Experiments with model of GMP compliant reproductive medicine laboratory confirm that the new method provides optimal manufacturing maintaining GMP and high product quality through the visual representation of the entire system and activity to bring into notice the deviations.

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1. Introduction

Good practices are applied to all the health care organizations as a systematic and comprehensive quality effort in laboratories, manufacturing set-ups and clinical centers for providing quality assurance (QA) as exemplified by Heibeler et al. (1988), Argote and Epple (1990), Vastag and Whybark (1991), Fitz-enz (1993), Adam (1994), Griffin et al. (1995), Lighter and Fair (2000), McCormick (2002), Ketokivi and Schroeder (2004), Carson and Dent (2007) and World Health Organization (2007). At times the organization is forcibly subjected to follow 'Good Manufacturing Practices'(GMP) regulatory guidelines advocated in literature viz., Laverty (1996), Waxman (2006), US FDA (2006) and Q8 ICH (2007). However, in all such cases, respective organizations own GMP set up because the respective management is supposed to have clear idea of the goals and comprehensions by the users (Benson et al., 1991; Diaz, 1994; Kenny, 1995; Brown et al., 2000; Jennings et al., 2000; Woodcock, 2004; Young and McClea, 2008). Now, according to Sharp (1983), Bailey (2001) and Paul and Tim (2001) any organization is not a uniform platform in terms of implementing all the GMP parameters into it rather an organization is a heterogeneous and anisotropic system resulting in the fact that, the GMP parameter performance pattern may not be uniform and those different parameters will have different contribution pattern to the final output.

For research and practical application, information regarding each of GMP sub-parameter's implementation patterns in specific section of the health care organization like contraceptive manufacturing organization would be of considerable value (Checkland, 1993; Bohn, 1994; Bongaarts and Bruce, 1995; Costello et al., 2001; Green, 2008). For example, in the process of achieving quality by parameter design above information would help in the correlation of product performance and customer satisfaction with the specific process design in the organization (Saraph et al., 1989; Thomas and Daley, 1996; Jha et al., 2010a). Simultaneously, this detailed and specific knowledge of each of GMP parameter's individual contribution pattern to the product of interest would help in case of cost reduction of health care products, improvement of

* Corresponding authors. Tel.: +91 3222 283574/75; fax: +91 3222 282221 (S.K. Guha).

E-mail addresses: jha.rk.pk@gmail.com, jha_pradeepk@yahoo.com (P.K. Jha), jharakhi@rediffmail.com (R. Jha), rajuldatt@rediffmail.com (R. Datt), guha_sk@yahoo.com (S.K. Guha).

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internal efficiency as well as, for competitive advantage of organization by setting of ideal process design and resource allocation for total utilization (Mandelbaum, 1990; Neuman et al., 1993; Flynn et al., 1994; Wiendahl and Scholtissek, 1994; Kelld and Sonnleitner, 1995; Bunney and Dale, 1996; Upton, 1996).

Sidney and Stoker (2001) emphasizes that during regulatory inspection of GMP parameters pertaining to any organization, the parameter performance in critical sections, for example, the premises, facility and process are of considerable interest. But Nair (2006) and Lande (2002) contradict that performance pattern of all the GMP parameters in correlation with product performance can't be measured readily. This is because in most cases, only the effect of single parameter that is, process variable in correlation with product characteristics is assessed (Spencer, 1994; Doblhoff-Dier and Bliem, 1999; Kourti, 2004). Therefore, considerable uncertainties and errors in interpretation may arise out of this procedure.

Recently, attempts have been made to measure the impact of various GMP parameters like environmental variable, which is associated with the process parameter variable that in turn affects the final product performance. This is done by adjusting the control factor of process variable at optimal level to achieve desired environmental change; the method named as "robust design" explained by Phadke (1989). This method too has its limitations in that measurement of some partial feature of a particular variable could be done (Pesch, 1994). Moreover, experiments on industrial product are possible (Pearson, 1933). Extrapolation to biological health care products leads to considerable inaccuracies due to difference in process complexities and clinical parameters (Laffel and Blumenthal, 1989; Mohammed et al., 2001).

Existing methods do not cover GMP and clinical parameters simultaneously (Dean and Bowen, 1994; Langley et al., 1996; Sole and Goodwin, 2000; Higgins, 2002; Andrew, 2009). For that, an alternative approach is presented in this paper to determine the totality of GMP parameter performance pattern by computation based on a model of the organization comprising all the GMP parameters. Keeping in view the percentage inefficiency in individual GMP parameter's implementation to the system, the totality of relationship can be addressed by the term 'entropy' (Georgescu - Roegen, 1971; Taschdjian, 1974; Haynes et al., 1980; Swanson et al., 1997; Qiu, 2002). Entropy is about order compared with disorder in a system (Kapur and Kesavan, 1992; Bailey, 1997; Baranger, 2001). From an information-theoretic perspective, entropy is defined as the amount of information required to describe the state of the system (Shannon, 1948). The use of entropy as a measure of organizational functionality assessment (Kumar, 1987; Frizelle, 1996; Bushuyev and Sochnev, 1999; Shuiabi et al., 2005; Martınez-Olvera, 2008) boost up the concept of our approach to visualize the GMP information and process parameter requirement interactions followed by entropy calculation with the histogram of that visualized system-process interaction. The limitations in terms of inaccuracies due to approximation are offset by readiness with which details of parameter performance pattern with changing environmental and production conditions can be obtained (Gary and Winbush, 1984). As an approach to address all these issues, this paper presents such a computational approach that looks for the totality of relationships between different variables inspite of relation between two individual entities.

2. Theoretical basis for good manufacturing practices conceptualization

In most variation/disorder investigations as well as in cases of side effect and complication in usage of health care products (Wennberg and Gittelsohn, 1973; Limpaphayom et al., 1997; Berwick, 1998), variable accumulation is commonly encountered during total time of batch production in given facility (Gosink et al., 2007); leading to displacement of variables as well as variation in product manufacturing steps (Ueda, 1993; Upton, 1995; Leon, 1998; Mukherjee et al., 1998; Park et al., 2001; Ting et al., 2009). Consequently, the variation during batch processing can be represented by an equivalent series resistance-inductance-capacitance (R-L-C) circuit (Odum, 1967) where the circuit parameters are frequency dependent. Here it is assumed that R-L-C circuit represents good manufacturing system where circuit parameters represent GMP parameters, frequency represents incoming process variable, which precede the activity of product development. Therefore, as first approximation for any given process range, designed GMP facility can be represented by simple resistance for variable. However, the magnitude of the resistance would be different from one process step to other process step. For the assessment of variable distribution in a given processing system, we can use the Ohm's law, that is

$$J = 1/\rho(E) = -1/\rho \text{ grad } V, \tag{1}$$

where J is the variable accumulation in facilitated processing environment, E is the process strength, V is GMP parameter performance at a point in the system and ρ is the process robustness. or by divergence theorem

$$\text{div } J = 0. \tag{2}$$

The statement assumes that there are no sources of variable inside the system as illustrated in Fig. 1. The manufacturing layout, environmental condition etc. are variable sources but the variation and imperfection imparted by these factors are negligible when compared with variables generated out of the process input. In quality assurance, the pattern of using variables in experimental design is so selected as to differentiate the system with the process input variables. Hence for analysis, variation estimated by design of experiment can be considered independent of the system with the process input variables. Since

$$\text{div}(-1/\rho \text{grad } V) = 0,$$

then in an anisotopic region

$$\frac{\delta}{\delta_x}\left(\frac{1}{\rho_x}\frac{\delta V}{\delta x}\right) + \frac{\delta}{\delta y}\left(\frac{1}{\rho_y}\frac{\delta V}{\delta y}\right) + \frac{\delta}{\delta z}\left(\frac{1}{\rho_z}\frac{\delta z}{\delta V}\right). \tag{3}$$

In an isotropic region where ρ is constant, this reduces to Laplace's equation

$$\nabla^2 V = 0. \tag{4}$$

A general solution of above equation is not possible because the GMP is too complex for mathematical description in its entirety (Simon, 1962; Albert et al., 2000; Azzouni, 2004). For that reason, simplification of assumption is necessary and the nature of the assumption depends on the totality of the GMP parameters to be correlated with product characteristics/users demand and the mode of variables introduced. For further analysis, variation distribution or entropy in the quality system of GMP parameter and process state will be considered. From the application point of view, the case has been found very important, particularly in relation to the reproductive medicine development (Guha et al., 1993; Guha, 1996; Guha et al., 1997; Jha et al., 2009a,b; Jha et al., 2010a,b). In a typical case, analogy wise V is the state variable representing process states and μ represents the GMP parameters. Process variables (V) in the form of V_A = Raw material input, V_G = Vacuum input, V_D = Heat input, V_H = Intermediate input, V_F = Gamma irradiation dose rate and total dose interrelation input are introduced in the quality system of GMP parameters (μ), that consist of μ_R = Premises, μ_P = Air particle, μ_s = Personnel, μ_Q = Training, μ_X = Process

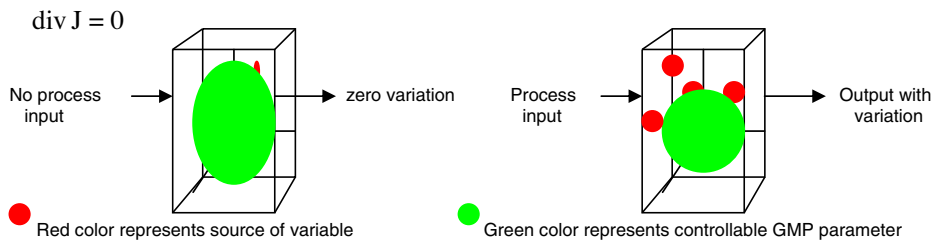

Fig. 1. Variation effect differentiation model.

validation, μ_W = Documentation, μ_U = Facility and equipments, μ_V = Packaging, where optimal condition of process variable and output quality will be V_O = Optimal and V_C = End product output. The system is suitably prepared as per GMP with optimizing variable spectrum to minimize the interface resistance.

In these circumstances, mechanism of variation distribution pattern would be interesting in the sense that all points of the interface of a specific quality system in contact with process variable will have variability equal to the variability of the respective process variables and at all other interfaces of the system the normal variability will be zero. The later condition is obtained because the system interface in contact with specific quality system interface has negligible source of variation, and the ratio of specific quality system variation to variation of connected system will be infinite (Fig. 2).

Within the good manufacturing system an array of inhomogenities exists. Some GMP parameters have anisotropic properties, for example, the variation depends on size of the working place (Frizelle and Woodcock, 1995; Baines et al., 2005). The variation permeable parameters of GMP sub-component particle count and process state are typical examples. Taking the quality system as a whole (Fig. 2) the process control, facility, equipment, premises, personnel, quality control, packaging and subunits, which due to their internal correlation lead to anisotropic properties, and so they are not aligned in parallel (Fig. 3). There do exist sufficient variation in alignment that the overall bulk homogeneity in different direction is similar. Therefore, gross variation distribution determinations and the GMP parameter sub-component air particle anisotropicity may be neglected without significant error and, may be taken to represent the resistivity at a point in all directions. Another significant inhomogeneity depicted by Taylor et al. (2000) is that, various GMP parameters in quality system have different deficiencies when process variables are introduced. The documentation of quality system procedure, for example, have a surveyed average deficiency of 8% whereas deficiencies associated with other parameters are contamination chemical/physical 7.3%, documentation–manufacturing 5.6%, handling and control of packaging component 5.5%, design and maintenance of premises 5.5%, in-process control and monitoring of production operation 4.9%, line clearance segregation and potential for mix-up 4.0%, process validation 3.5%, design and maintenance of equipment 3.4%, environmental monitoring 3.4%, supplier and contractor audits and technical agreements 3.4%, calibration of measuring and test equipment 3.3%, equipment validation 3.1%, starting material and packaging component testing 2.8%, sampling- procedure and facility 2.7%, status labeling – work in progress, facility and equipment 2.7%, housekeeping 2.2%, warehousing and distribution activities 1.9% and personnel issues-duties of key personnel 1.7%. Additional complexity arises because quality system is not just personnel training and qualification; controlling the product design; controlling documentation; controlling purchasing; product identification and traceability at all stages of production; controlling and defining production and process; defining and controlling inspection, measuring and test equipment; validating processes; product acceptance; controlling nonconforming product; instituting corrective and preventive action when errors occur; labeling and packaging controls; handling, storage, distribution and installation; records; servicing and quality assurance techniques; but quality system also involve management practices, manufacturing practices, operation practices and strategy to build the quality into the product.

Consideration of all these inhomogeneities is not possible at this stage (Goldenfeld and Kadanoff, 1999). Appropriate simplifying approximations are that the quality system, which includes personnel/premises, process control/qualification, sterile production and

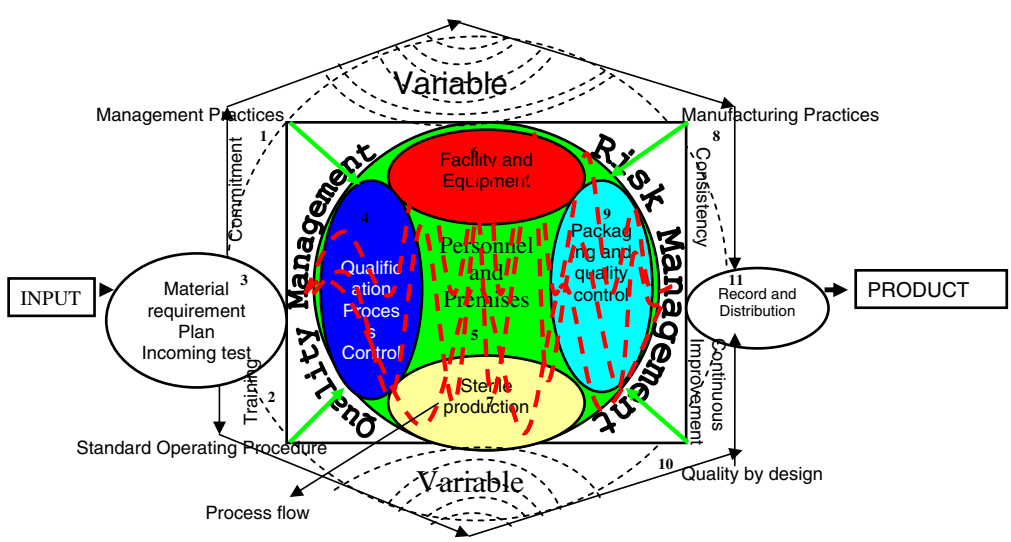

Fig. 2. Model for good manufacturing system where process variable interact with GMP variable.

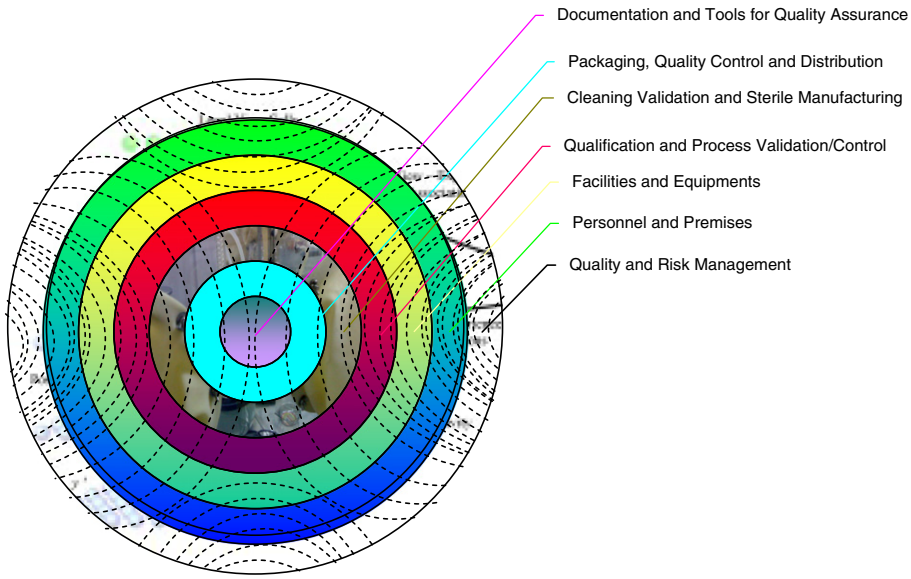

Fig. 3. Variation distribution pattern in good manufacturing practices.

quality control/packaging having a definite correlation and distinct interfacial boundry (Deshmukh et al., 1998; ElMaraghy and Urbanic, 2004). Also, the quality system is permeable to process variable with definite in process control deficiencies 4.9% as per GMP inspection reported by Taylor et al. (2000). Since the inadequecy of a quality system is between that of the process and the environment (premises), and at the same time system operator (personnel) is situated between the process and working environment in the quality system; the effect of variation distribution may be taken into account by increasing the robustness of process and considering the quality system environment to be continously checked. On the basis of actual experiment in relation to process and controlled quality system used for reproductive medicine manufacturing; the robustness of process variable may be considered as optimal for the present experimentation.

The outer boundary of the quality system as well as the interface between internal component of quality system are found to be parabola shaped and process stepped irregularly in wave form as shown in Fig. 2. It elaborates that the outer boundary is based on management practices (Dale and Lightburn, 1992), manufacturing practices, quality building approach through process and standard operating procedure (SOP) while its funcationalization is based on the strategy of management commitment, process consistency, continous improvement and training method. Wave form of process variable interaction with system can be considered for experimental analysis. For quality problem, a more appropriate method is to approximate the widespread curved system variables as explained in Fig. 3 and waved form of process movement shown in Fig. 2 by stepped straight lines. The steps are selected in such a manner as to have rough number of corner nodes inside and outside the process movement curve. With such a selection following computation with the straight line boundaries, the deviation on curved part of the process step can be calculated by interpolation taking into account the crossing point of the process variable wave curved with the straight line approximation.

The quality system for GMP is a three-dimensional mesh structure with permeable variables throughout the system. As a result, even if process variable is introduced at the processing stage inside the quality system of GMP, variability may spread to all work areas that consist of interconnected processes until a boundary with process control encountered (Sulgrove and Herndon, 1993); but the variation accumulation rapidly declines as one control the process at that point and process moves away from the variable introducing axis in the Z direction (Documentation to personnel issues). Therefore, in the Z axis, only a flow of process that would effect the quality system need to be taken into account, and interface environmental area where the process to be moved may be assumed to exist beyond this process flow. A method for assessing the effect of that flow of process, which induces the variability distribution in the good manufacturing system has been proposed and applied.

3. Analysis of conceptualized model

With the approximations made, the GMP system component for analysis in the X, Y plane is shown in Fig. 2. In the Z direction, a similar layout exists. Since the system efficiency against variation accumulation in each individual component of the system is assumed to be uniform, the variation distribution for an individual GMP component is determined from Eq. (5). In order to apply this equation to the entire system, the solution at the process interfaces must be properly matched. It can be achieved by clearly written SOP. The continuity conditions were that the tangential iteration of the process and system efficiency on the different sides must be similar. Within each homogeneous part the standard Laplace equation will be valid. Analytical solution of the Laplace equation for the complicated system like GMP revealed in Fig. 2 is not feasible. An alternative is to use an experimental method. To compute the variation pattern optimally as given away in Fig. 2, system was divided into three-dimensional meshes like the one shown in Fig. 4. For convenience, the mesh was adjusted so that all the interface of GMP parameter boundaries within the quality system passes along mesh lines and faces. The GMP format implementation is considered to be uniform. The total number of nodes in one layer was 479, considered a node (i,j,k). If the node is not a boundary node, by Taylor's series expansion, the Laplace equation at node (i,j,k) could be represented in the form (Vitkovitch, 1966).

$$V_{(i+1,j,k)} + V_{(i,j+1,k)} + V_{(i-1,j,k)} + V_{(i,j-1,k)} + V_{(i,j,k+1)} + V_{(i,j,k-1)} - 6V_{(i,j,k)} = 0, \quad (5)$$

where i, j, k are the coordinate indices in the X, Y and Z directions, respectively.

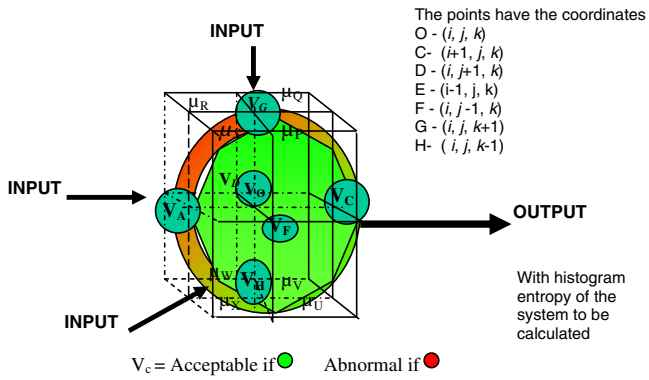

Fig. 4. Three-dimensional GMP parameter representations (Location of GMP parameter in three-dimensional format and interaction with process state variable).

Equations at boundary nodes of different types have been worked out and reported in the literature for the two-dimensional case. For GMP quality system with the large number of nodes and boundaries, a generalized form suitable for generating the equations conveniently in three-dimensions is required. Based on the continuity conditions mentioned earlier a generalized equation has been derived for a node O (i, j, k) having all eight quadrants of different GMP parameter, as shown in Fig. 4. The generalized equation is

$$\begin{aligned}&(\mu_U+\mu_V+\mu_P+\mu_Q)V_{(i+1,j,k)}+(\mu_V+\mu_Q+\mu_R+\mu_w)V_{(i,j+1,k)}\\&\quad+(\mu_X+\mu_W+\mu_R+\mu_S)V_{(i-1,j,k)}+(\mu_X+\mu_U+\mu_P+\mu_S)V_{(i,j-1,k)}\\&\quad+(\mu_p+\mu_Q+\mu_R+\mu_S)V_{(i,j,k+1)}+(\mu_U+\mu_V+\mu_W+\mu_X)V_{(i,j,k-1)}\\&\quad-3(\mu_P+\mu_Q+\mu_R+\mu_S+\mu_U+\mu_V+\mu_W+\mu_X)V_{(i,j,k)}=0,\end{aligned}\tag{6}$$

where μ represents the GMP parameter of the quadrant specified by the subscript. The equation was also applicable to surface points if the quadrants representing parameter outside the manufacturing process parameter were assigned a probability of variation negligible such as packaging control and inventory control.

The accumulative effect (A_s) of GMP parameter (μ) distributed at the eight different quadrant of quality system (see Fig. 4) is considered to be determined by Eq. (8).

$$\mu_{\text{average}}=\frac{\mu_R+\mu_S+\mu_Q+\mu_P+\mu_W+\mu_X+\mu_V+\mu_U}{8},\tag{7}$$

$$A_s=k\times\mu_{\text{average}}.\tag{8}$$

The variation accumulation at a point can be computed by representing Eq. (1) in its numerical form by applying equation explained in Fig. 5. Thus

$$J_x=-\frac{1}{\rho}\frac{\delta V}{\delta x}=-\frac{1}{\rho}\left(\frac{V_{(i+1,j,k)}-V_{(i-1,j,k)}}{2h}\right),\tag{9}$$

$$J_y=-\frac{1}{\rho}\frac{\delta V}{\delta y}=-\frac{1}{\rho}\left(\frac{V_{(i,j+1,k)}-V_{(i-1,j,k)}}{2h}\right),\tag{10}$$

$$J_z=-\frac{1}{\rho}\frac{\delta V}{\delta z}=-\frac{1}{\rho}\left(\frac{V_{(i,j+1,k)}-V_{(i-1,j,k)}}{2h}\right),\tag{11}$$

where h is the mesh interval. The resultant variation accumulation is

$$J=\left(J_x^2+J_y^2+J_z^2\right)^{1/2}.\tag{12}$$

The variable GMP parameters were considered to be of square form and covering two mesh intervals. Analysis for parameter of one mesh interval square (i.e. a surface of h^2 was also performed. Appropriate GMP in quality system for analysis were obtained by assigning to the nodes in contact with the process variable at one side of the quality system, a variability of zero and on the other side a variability of one percent.

Generally, the formulation of difference equations for each node, based on Eq. (6) does not present any particular problem. Standard solution programs are also available. But the present problem is of unusually large size because a single plane involves 479 nodes and the equation for each node involves six other nodes. Therefore, special techniques for economy in time and computer memory are required to make a solution practicable. The conventional method is to represent the values of process capacity in the equations as subscripted variables.

The method used here was to write the equations of entropy with specified coefficients and to incorporate them in the main computer program as arithmetic statements. For the large number of equations involved, writing the equations in proper form manually was in itself a large task. Therefore, another computer program was written to generate the equations in the proper form. The data input to this program were the eight appropriate quadrant GMP parameter index values for each node. The output was in the form of difference equations, which were stored for use in the main program.

The set of simultaneous equations were solved by Gauss Seidel iteration on Matlab 7.0 with Young's point relaxation (Young, 1954) for accelerating convergence. Computations were carried out to accuracy, where the node deviations after a given iteration were less than 0.0001% different from the values of the previous iteration. This accuracy was achieved in around 300 iterations using a relaxation factor of 1.7. The processing was done on an Intel (R) Duel core 2.5 GHz computer with a 4 GB RAM. The computer analyses were performed first for a single median plane with 479 nodes as if the problem was two-dimensional. Next, a three-dimensional model of three planes with 1437 nodes was analyzed. Finally, another three-dimensional problem with five planes and 2395 nodes was solved derived in Fig. 5.

In Fig. 5, red color[1] represent the source of variation having negligible impact on the performance of all individual GMP parameter (see in Fig. 4) accumulated (see Eq. (8)), while green color correspond to the wellness of good manufacturing system. On the whole, the visualized GMP system entropy is calculated with the histogram.

4. Experimental verification of analysis with model of pilot laboratory

In order to verify the trends predicted by theoretical analysis, an experimental model has been used in this paper. The model was a pilot laboratory as shown in Fig. 6. The laboratory was of GMP compliance having controlled environment maintaining a particle count matching with that of the industrial manufacturing set-up. The process was simulated by gamma irradiation dose rate and total dose of matching variability (Jha et al., 2010a). The processing area was modelled with acrylic made square shape and the process with variability of gamma dose rate, vacuum and temperature. Experiments were performed with people having variable year of understanding like $1\times1/2$, 1×2 and 1×3 years about process system uncertainty, connected to one percent variance SOP and 10 year documented record source. Entropy or deviation was measured by the visual perception followed by image processing method at various points in the simulated process and inside the processing area. In order to obtain variation densities at each particular process point, people were employed with 5 day interval of completion of one process cycle and deviation calculation.

[1] For interpretation of colour in Fig. 5, the reader is referred to the web version of this article.

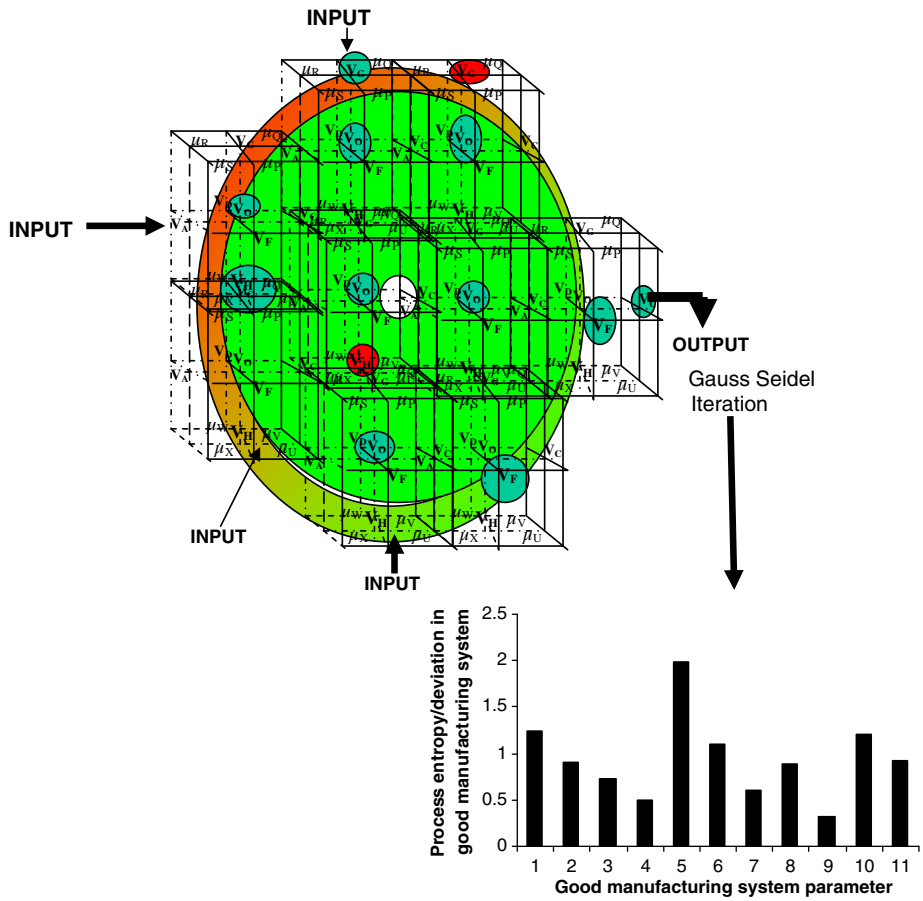

Fig. 5. Combination of all GMP parameters with all process variables maintaining optimal GMP condition at minimum entropy. Entropy calculation with histogram visualizes the model.

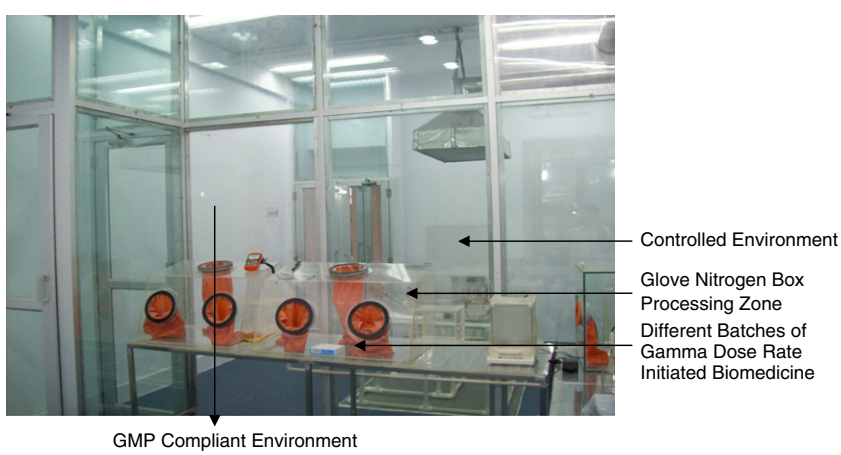

Fig. 6. Visualized laboratory model for practical system process interaction and entropy/deviation calculation.

5. Results and discussion

Results of the computerization indicate that the general pattern of variation distribution lines in the median plane remains unaltered by the addition of the other planes into the analysis. Taking a point within the processing, the absolute value of deviation changes by approximately 10% on the addition of two GMP parameter planes and by a further 7.1% on addition of two more planes. Therefore, with Z direction homogeneity, it is adequate to consider three planes or the GMP parameter actually go through by the people with variable process understanding (Harrington, 1984). The variation distribution lines for such a situation in the median plane are shown in Fig. 2, deviation accumulation at the significant points marked are given in Table 1.

As per the above observation, calculations and their analysis; variation distribution lines near the people in processing zone are closely spaced, whereas in the central part of the system the lines are more parallel and spaced apart. A similar situation results in the Z direction. This indicates that the variation accumulation is high near the people having casual concept of process and

Table 1
Details of entropy or deviation value calculated for different areas of GMP model.

Area code point [See Fig. 2]	Standard deviation (in terms of process flow inside the processing environment)
1	1.25
2	0.91
3	0.73
4	0.50
5	1.98
6	1.10
7	0.61
8	0.89
9	0.32
10	1.20
11	0.93

decreases in the central region. A practical consequence of this fact would be that in actual simulation of the good manufacturing system, there is a possibility of maximum side effect, complications creating product near the people having very less understanding of process implication within the good manufacturing system. Therefore, optimal process understanding is desirable for totality of GMP compliance, consistency and reproducibility.

Considering deviation accumulation and the variability it can be inferred that a significant part of people-process interaction passes through the system. When calculations were repeated for the people having small process interaction, it was found that the "intelligentsia" involvement into deeper process of product development was poorer (Dretske, 1999). For GMP compliant purposes, maximum people-process interaction would be desirable because a larger proportion of the intelligentsia would pass through the product quality stabilization practices (Kolb, 1984; Schon, 1988). On the other hand, for accidental interaction with variation sources, it is to be observed that GMP ineffectiveness due to variable is more likely if the interaction occurs within large variably fluctuating sources like process and people intelligentsia, than within small fluctuating sources like system.

In the experimental observations, the variation distribution pattern obtained broadly corresponded to the theoretical result. For different people having different process interaction, when the deviation accumulation were corrected by a factor corresponding to the change in years of total people-process interaction due to process flow change, it was observed that intelligentsia usability improved with increasing people-process interaction regarding their implication on final product quality. This way the experimental results show trends similar to the theoretical analysis results.

6. Conclusion

It may therefore be concluded that this method of computation leads to a picture of the variation distribution pattern or entropy inside the good manufacturing system. The technique can be readily adapted for other components of good practices like good laboratory practices (GLP), good clinical practices (GCP) and so on. This paper identify entropy or variation distribution trend and provide insight into regions of interest through coherent and meaningful visualization. This explains that how sets of variables in complex datasets interact throughout regions of interest and the role other variables play in influencing these interactions as well.

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